

Executive Assessment Framework: Plumbing Contractor Selection

Project Identification: Second-Floor Alcove Bathtub Installation (~80 lbs)

Location: Canyon Lake, TX

Site Parameters: * Open studs/joists

- Original builder rough-in
- Private septic system
- **Zero first-floor ceiling access** (Top-down connection required)
- Partial plywood subfloor intact at install site
- Shower mixing valve required (Hardware pending)

Phase 1: Mandatory Prerequisites (Pass/Fail)

Evaluation stops if the candidate fails to provide verified documentation for the following risk-mitigation criteria.

- ☐ **State Licensing Verification:** Active Journeyman or Master Plumber license via the Texas State Board of Plumbing Examiners (TSBPE).
- ☐ **High-Limit Liability Insurance:** Verified Certificate of Insurance (COI) explicitly covering structural and lower-level water damage.
- ☐ **Workmanship Warranty:** Written guarantee on labor and connection integrity (minimum 1–2 years), independent of manufacturer fixture warranties.

Phase 2: Weighted Evaluation Matrix

Scoring Scale:

- **5 – Exceptional:** Industry-leading capability, proactive problem solver, highly technical.
- **4 – Strong:** Above-average performance, solid technical answers.
- **3 – Adequate:** Meets basic expectations safely.
- **2 – Weak:** Noticeable gaps in technical knowledge or site preparation.
- **1 – Unacceptable:** Immediate disqualification; proposes dangerous or non-compliant practices.

Core Competencies	Weight	Score (1-5)	Weighted Score
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1. Blind-Space & Multi-Story Expertise25%

(Execution of top-down connections without ceiling access)

2. Structural Support Execution20%

(Load dispersion and securing protocols for lightweight tubs)

3. Layout Precision & Troubleshooting20%

(Adapting legacy rough-ins to specific fixture tolerances)

4. Local Code & Septic Competence15%

(Comal County venting compliance and septic-safe practices)

5. Pricing Transparency & Management20%

(Itemized quoting, change-order protocols, scheduling)

TOTAL SCORE100%—/ 5.0

(Formula: Weighted Score = [Score ÷ 5] × Weight. Example: A score of 4 on Layout = (4/5) * 20 = 0.16)

Phase 3: Technical Interview Protocol

Conduct this interview on-site at the second-floor installation area. Evaluate responses against the 1-5 scale above.

1. The Blind Connection Test (Logistics & Code)

Prompt: "The downstairs ceiling is fully finished and will not be penetrated. How will you secure the drain and P-trap entirely from above, and what specific fittings will you use?"

- **Target (Score 5):** Details a top-down approach within the joist bay using **solvent-welded (glued) joints**. Acknowledges that threaded "slip-nut" fittings are code-prohibited in permanently inaccessible concealed spaces.

- **Red Flag (Score 1):** Insists on cutting the downstairs ceiling or proposes using slip-nuts in a concealed cavity.

2. The Structural Support Test (Layout Best Practices)

Prompt: "This is an 80-pound alcove tub. What is your protocol for securing the base to prevent flex, squeaking, or flange detachment under load?"

- **Target (Score 5):** Mandates two elements: 1) A horizontal **ledger board (stringer)** secured across the wall studs. 2) A **mortar bed** (or Structolite/plaster pad) on the subfloor to eliminate void space and lock the tub base.
- **Red Flag (Score 1):** Suggests screwing the flange to the studs and resting the tub directly on the bare plywood subfloor.

3. The Legacy Rough-In Test (Troubleshooting)

Prompt: "The original builder installed this rough-in. If the drain placement is off-center or obstructed by a joist relative to the new tub spec, what is the corrective action?"

- **Target (Score 5):** Takes immediate measurements. Requests the tub specification sheet. Proposes using solid PVC offset fittings to route the drain safely within the joist bay without compromising structural lumber.
- **Red Flag (Score 1):** Proposes notching the top of the load-bearing floor joist or using flexible corrugated piping (illegal and high-risk).

4. The Septic Venting Test (Zoning & Local Code)

Prompt: "We operate on a private septic system. How will you verify the existing builder rough-in provides adequate venting to prevent sewer gas backflow?"

- **Target (Score 5):** Visually traces the vent stack. Confirms the P-trap size and distance-to-vent ratio complies with Texas code to prevent trap-siphoning under the high velocity of a 40+ gallon gravity drain.
- **Red Flag (Score 1):** Assumes the builder's work is adequate without visual verification.

5. The Leak Prevention Test (Risk Mitigation)

Prompt: "Prior to closing the subfloor and installing cement board, what is your testing protocol to guarantee zero leakage onto the finished first-floor ceiling?"

- **Target (Score 5):** Requires a **static water test** (plugging the drain, filling to the overflow line, and draining) to stress-test joints under maximum flow/weight before covering the work.
- **Red Flag (Score 1):** Proposes running the water briefly to check for drips.

Phase 4: Final Contract Checklist

Prior to execution, the finalized contract/SOW must explicitly include the following line items:

- ☐ **Mortar Bed Installation:** Explicitly listed in the plumbing scope of work (not deferred to the tile contractor).
- ☐ **Ledger Board / Stringer Installation:** Included in structural prep.
- ☐ **Rough-In Valve Specifications:** Written acknowledgment of the exact shower mixing valve brand/model, including the calculated mounting depth based on the planned wall finish thickness (e.g., 1/2" cement board + thinset + 3/8" tile).
- ☐ **Subfloor Reinforcement:** Agreement to install structural blocking (scrap 2x4s) beneath the edges of any subfloor cutout made for drain access, ensuring the mortar bed rests on a fully supported surface.
- ☐ **Static Water Test:** Defined as a required milestone before framing/drywall contractors are permitted to enclose the plumbing.
- ☐ **Division of Materials:** Clear delineation of who supplies the tub, drain/overflow trim kit, and shower mixing valve.